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(54) **BOARD-LEVEL PAD PATTERN FOR  
MULTI-ROW QFN PACKAGES**

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See application file for complete search history.

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(57) **ABSTRACT**

A board-level pad pattern includes staggered ball pads  
disposed within a surface mount region for mounting a  
multi-row QFN package. The staggered ball pads include  
first ball pads arranged in a first row and second ball pads  
arranged in a second row. The first ball pads in the first row  
are arranged at two different pitches, and the second ball  
pads in the second row are arranged at a constant pitch.

**16 Claims, 3 Drawing Sheets**

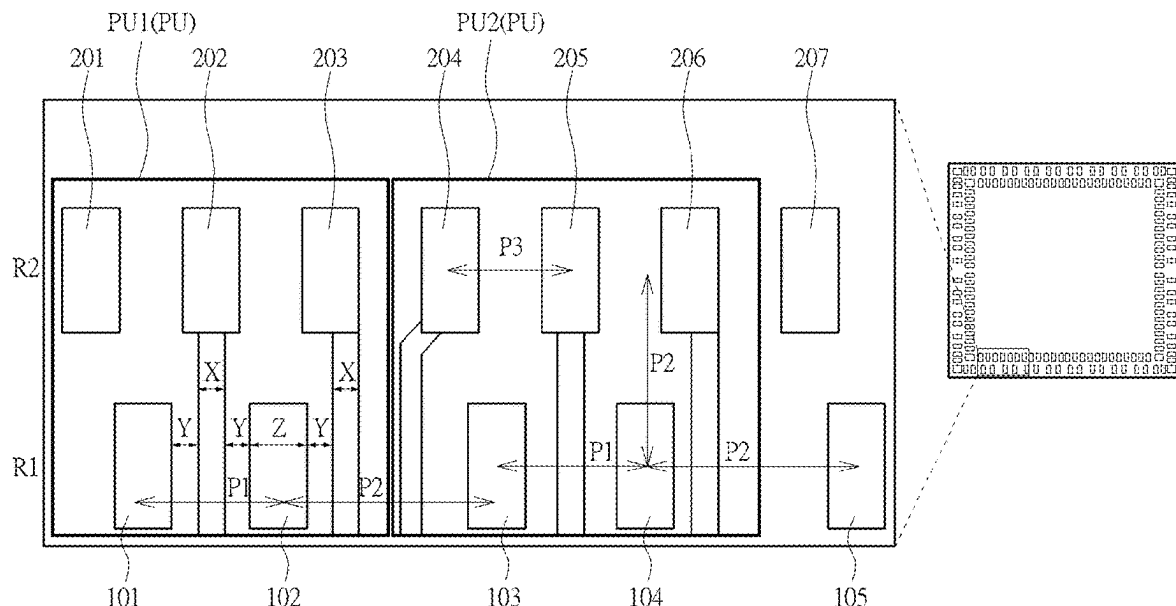
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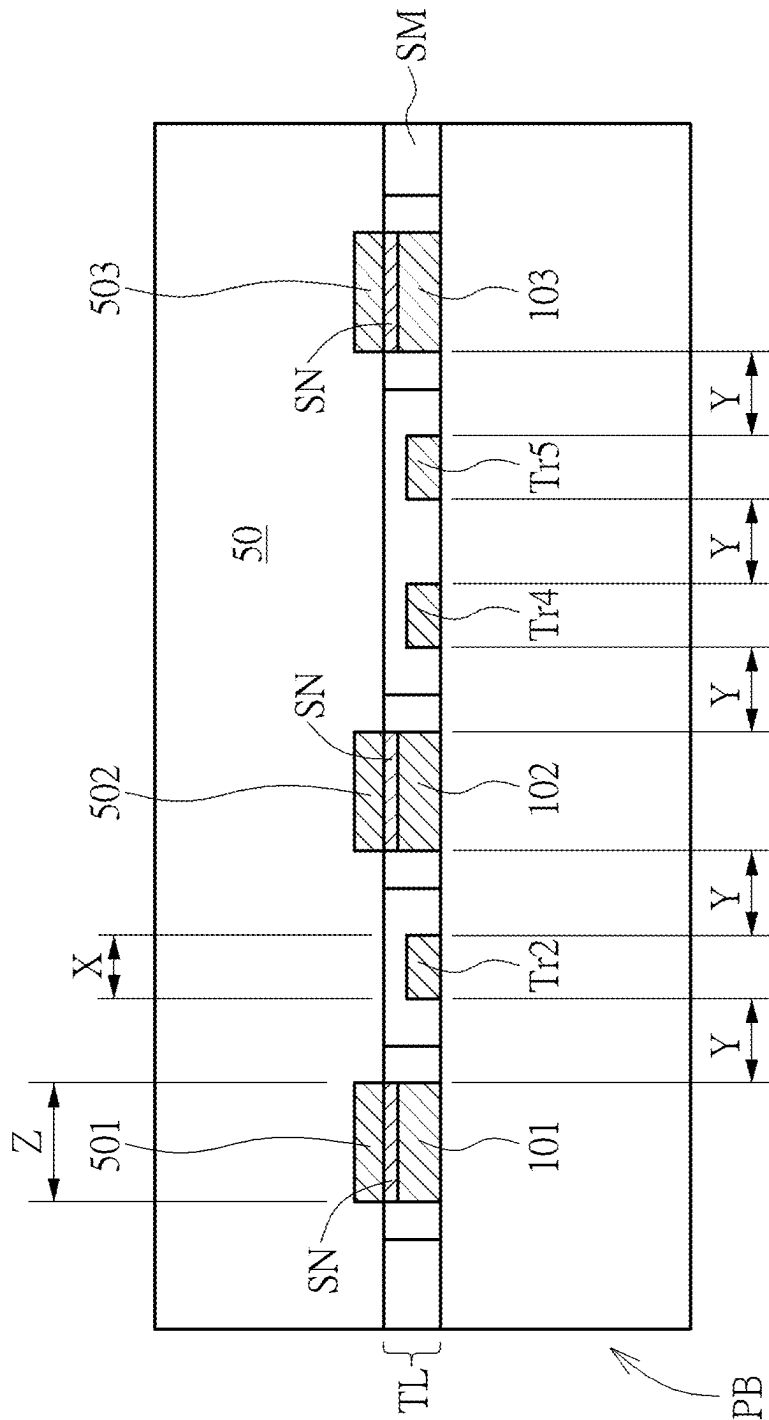


FIG. 2

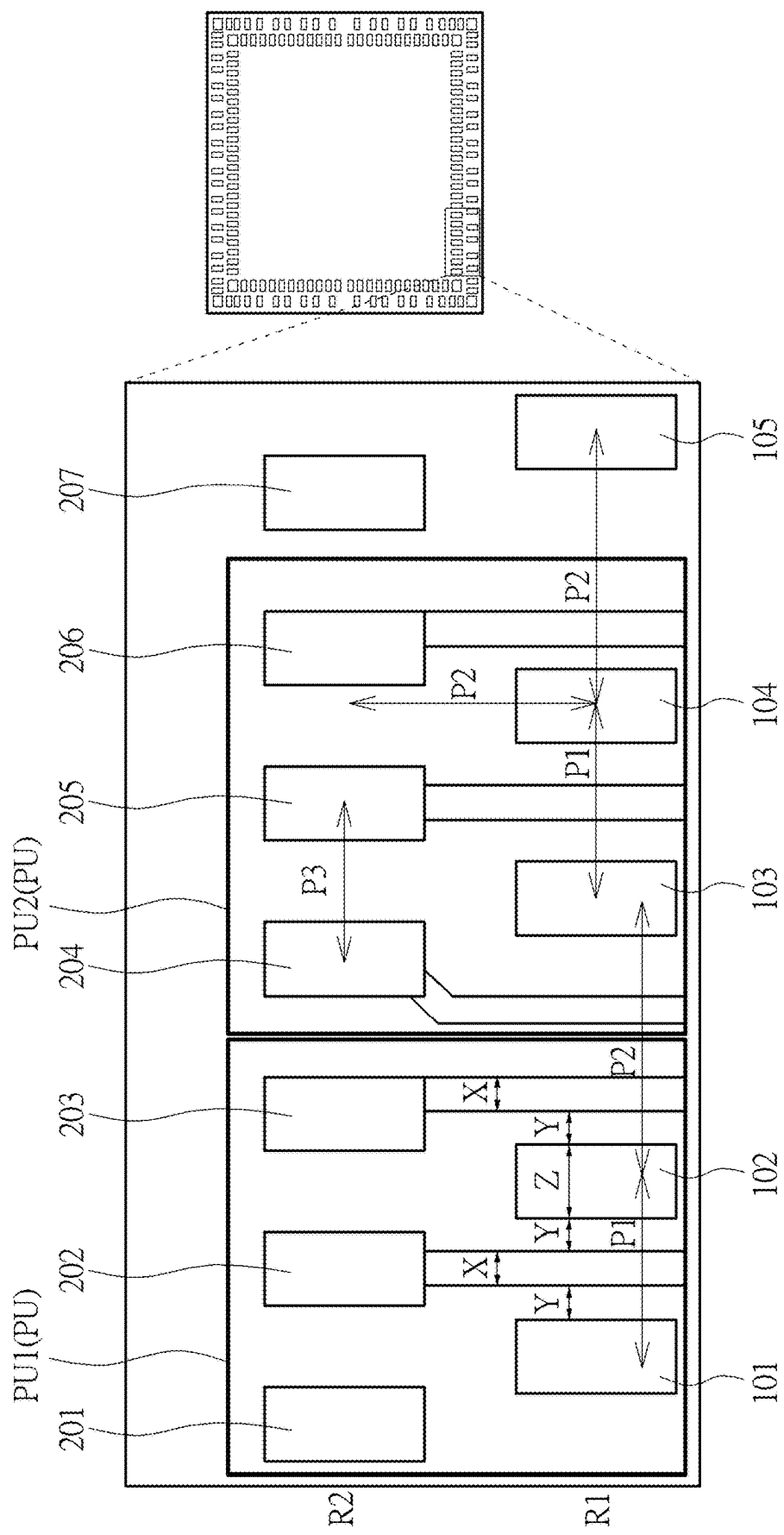


FIG. 3

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## BOARD-LEVEL PAD PATTERN FOR MULTI-ROW QFN PACKAGES

### CROSS REFERENCE TO RELATED APPLICATIONS

This application claims the benefit of U.S. Provisional Application No. 63/296,011, filed on Jan. 3, 2022. The content of the application is incorporated herein by reference.

### BACKGROUND

The present disclosure relates to the printed circuit board (PCB) ball pad (or finger pad) layout. More particularly, the present disclosure relates to a board-level ball pad pattern with differential pair routing for multi-row Quad Flat No lead (QFN) packages.

The continuous advancement in technology and miniaturization of electronic components, hand held and communication devices require superior thermal-electrical performance and miniature packages. An advanced and complicated Integrated Circuit (IC) device often demands increase in number of I/O's while maintaining its small size, footprint and weight. A dual-row QFN is an ideal solution for such demanding applications.

The multi-row QFN package is a plastic-encapsulated package with a copper leadframe substrate. The exposed die attach paddle on the bottom efficiently conducts heat to the PCB and provides a stable ground through down bonds or by electrical connections through conductive die attach material. The design of multi-row QFN packages allows for flexibility and enhances electrical performance to very high-speed operating frequencies.

As known in the art, differential signaling has been employed on PCB routing for the multi-row QFN packages, which uses two complementary signals to transmit the one data signal, but the second signal is inverted from the first. The signal receiver uses the difference between the inverted and non-inverted signals to decipher the information. Using differential pair routing to transmit signals has benefits such as reduction in noise and EMI. However, the drawback is that it takes up more space.

Further, to get the best performance out of the differential pair routing, the lines must be equal in length and the pairs must be routed together and at the same widths, which can be a problem when routing around obstacles on the circuit board such as vias or smaller components.

### SUMMARY

One object of the present invention is to provide an improved board-level ball pad pattern with differential pair routing for multi-row QFN packages that is capable of either reducing package size (same pin count) or increasing pin count (same package size) in order to solve the above-mentioned prior art problems or shortcomings.

One aspect of the invention provides a board-level pad pattern including a plurality of staggered ball pads disposed within a surface mount region for mounting a multi-row QFN package. The plurality of staggered ball pads comprises first ball pads arranged in a first row and second ball pads arranged in a second row. The first ball pads in the first row are arranged at two different pitches, and the second ball pads in the second row are arranged at a constant pitch.

According to some embodiments, the plurality of staggered ball pads is copper pads.

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According to some embodiments, the plurality of staggered ball pads is non-solder mask defined (NSMD) pads.

According to some embodiments, the first ball pads in the first row are aligned to one another along a first direction.

According to some embodiments, the second ball pads in the second row are aligned to one another along the first direction

According to some embodiments, the first and second ball pads are arranged in a staggered manner in a second direction that is orthogonal to the first direction

According to some embodiments, the first ball pads in the first row are arranged at a pitch of 0.5 mm and a pitch of 0.7 mm.

According to some embodiments, the second ball pads in the second row are arranged at a pitch of 0.4 mm.

According to some embodiments, the board-level pad pattern further comprises a basic pad unit consisting of five pads composed of two of the first ball pads in the first row and three of the second ball pads in the second row.

According to some embodiments, an array of the plurality of staggered ball pads within the surface mount region for the multi-row QFN is constructed by repeating the basic pad units.

Another aspect of the invention provides a board-level pad pattern including a plurality of staggered ball pads disposed within a surface mount region for mounting a multi-row QFN package, wherein the plurality of staggered ball pads comprises first ball pads arranged in a first row and second ball pads arranged in a second row, wherein the first ball pads in the first row are arranged at two different pitches P1 and P2, and the second ball pads in the second row are arranged at a constant pitch P3. At least two differential pair traces disposed between adjacent two of the first ball pads arranged in a first row.

According to some embodiments, the plurality of staggered ball pads is copper pads.

According to some embodiments, the plurality of staggered ball pads is non-solder mask defined (NSMD) pads.

According to some embodiments, the first ball pads in the first row are aligned to one another along a first direction.

According to some embodiments, the second ball pads in the second row are aligned to one another along the first direction

According to some embodiments, the first and second ball pads are arranged in a staggered manner in a second direction that is orthogonal to the first direction

According to some embodiments,  $P1 = X + 2Y + Z$  and  $P2 = 2X + 3Y + Z$ , wherein X is a width of each of the differential pair traces, Y is the space between the differential pair traces and the adjacent two of the first ball pads arranged in a first row, and Z is a width of each of the first ball pads.

According to some embodiments,  $P1 = 0.5$  mm,  $P2 = 0.7$  mm, and  $P3 = 0.4$  mm.

According to some embodiments, the board-level pad pattern further includes a basic pad unit consisting of five pads composed of two of the first ball pads in the first row and three of the second ball pads in the second row.

According to some embodiments, an array of the plurality of staggered ball pads within the surface mount region for the multi-row QFN is constructed by repeating the basic pad units.

Another aspect of the invention provides a semiconductor device including a base substrate comprising a plurality of staggered ball pads disposed within a surface mount region for mounting a multi-row QFN package. The plurality of staggered ball pads comprises first ball pads arranged in a first row and second ball pads arranged in a second row. The

first ball pads in the first row are arranged at two different pitches, and the second ball pads in the second row are arranged at a constant pitch. A basic pad unit consists of five pads composed of two of the first ball pads in the first row and three of the second ball pads in the second row. An array of the plurality of staggered ball pads within the surface mount region for the multi-row QFN is constructed by repeating the basic pad units.

These and other objectives of the present invention will no doubt become obvious to those of ordinary skill in the art after reading the following detailed description of the preferred embodiment that is illustrated in the various figures and drawings.

### BRIEF DESCRIPTION OF THE DRAWINGS

The accompanying drawings are included to provide a further understanding of the invention and are incorporated in and constitute a part of this specification. The drawings illustrate embodiments of the invention and, together with the description, serve to explain the principles of the invention. In the drawings:

FIG. 1 shows a germane portion of an exemplary basetop layer of a layout of ball pads for a multi-row QFN package according to one embodiment of the invention;

FIG. 2 is a schematic, cross-sectional diagram taken along line I-I' in FIG. 1 after the assembly of the multi-row QFN package; and

FIG. 3 schematically shows the partial ball pad layout for a multi-row QFN package and the pad pitches according to an embodiment of the invention.

### DETAILED DESCRIPTION

In the following detailed description of embodiments of the invention, reference is made to the accompanying drawings, which form a part hereof, and in which is shown by way of illustration specific preferred embodiments in which the disclosure may be practiced.

These embodiments are described in sufficient detail to enable those skilled in the art to practice them, and it is to be understood that other embodiments may be utilized and that mechanical, chemical, electrical, and procedural changes may be made without departing from the spirit and scope of the present disclosure. The following detailed description is, therefore, not to be taken in a limiting sense, and the scope of embodiments of the present invention is defined only by the appended claims.

It will be understood that when an element or layer is referred to as being “on”, “connected to” or “coupled to” another element or layer, it can be directly on, connected or coupled to the other element or layer or intervening elements or layers may be present. In contrast, when an element is referred to as being “directly on”, “directly connected to” or “directly coupled to” another element or layer, there are no intervening elements or layers present. Like numbers refer to like elements throughout. As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items.

FIG. 1 is a schematic diagram showing a germane portion of an exemplary base top layer of a layout of ball pads for a multi-row QFN package according to one embodiment of the invention. FIG. 2 is a schematic, cross-sectional diagram taken along line I-I' in FIG. 1 after the assembly of the multi-row QFN package. As shown in FIG. 1 and FIG. 2, the ball pad layout 1 fabricated in the top layer TL of a base substrate PB comprises staggered ball pads (or finger pads)

including, for example, ball pads 101-103 arranged in the first row R1 and ball pads 201-205 arranged in the second row R2. According to an embodiment of the invention, for example, the ball pads 101-103 and 201-205 may be copper pads, but is not limited thereto. The base substrate PB may comprise a printed circuit board or a package substrate, but is not limited thereto.

According to an embodiment of the invention, the ball pads 101-103 and 201-205 are disposed within a surface mount region 10 for mounting a multi-row QFN package 50 (FIG. 2) using surface mount techniques known in the art. The leads 501-503 on the bottom side of the multi-row QFN package 50 may be bonded to corresponding ball pads 101-103 and 201-205 with solder joints SN to form electrical connection paths. According to an embodiment of the invention, for example, the ball pads 101-103 and 201-205 may be non-solder mask defined (NSMD) pads. For the sake of simplicity, the solder mask is omitted in FIG. 1.

According to an embodiment of the invention, the ball pads 101-103 arranged in the first row R1 are approximately aligned to one another along the first direction D1. According to an embodiment of the invention, the ball pads 201-205 arranged in the second row R2 are approximately aligned to one another along the first direction D1. According to an embodiment of the invention, the ball pads 101-103 and 201-205 may be arranged in a staggered manner in the second direction D2 that is orthogonal to the first direction D1.

According to an embodiment of the invention, for example, the ball pad 101 may be connected to a trace Tr1, the ball pad 202 may be connected to a trace Tr2, the ball pad 102 may be connected to a trace Tr3, the ball pad 203 may be connected to a trace Tr4, the ball pad 204 may be connected to a trace Tr5, and the ball pad 103 may be connected to a trace Tr6. According to an embodiment of the invention, for example, the traces Tr1-Tr6 may extend along the second direction D2.

According to an embodiment of the invention, for example, only the trace Tr2 passes through the routing space between the ball pad 101 and ball pad 102 in the first row R1. According to an embodiment of the invention, for example, the two adjacent traces Tr4 and Tr5 pass through the routing space between the ball pad 102 and ball pad 103 in the first row R1. According to an embodiment of the invention, for example, the traces Tr4 and Tr5 are designated as a differential pair, and the adjacent balls pads 103 and 103 are designated as ground pads, thereby constituting a ground-signal-signal-ground (GSSG) routing configuration. In FIG. 1, the width of each of the ball pads 101-103 and 201-205 is labeled as Z, the width of each of the traces Tr1-Tr6 is labeled as X, and the space between the passing traces Tr2, Tr4 and Tr5 and the ball pads 101-103 is labeled as Y.

FIG. 3 schematically shows the partial layout of the ball pads and the pad pitches according to an embodiment of the invention. For the sake of simplicity, the solder mask and the traces are omitted in FIG. 3. As shown in FIG. 3, according to an embodiment of the invention, the ball pads 101-105 in the first row R1 are arranged at two different pitches P1 and P2, for example,  $P1=0.5\text{ mm}$  and  $P2=0.7\text{ mm}$ . According to an embodiment of the invention,  $P1=X+2Y+Z$  and  $P2=2X+3Y+Z$ . According to an embodiment of the invention, for example, the ball pads 201-205 in the second row R2 are arranged at a constant pitch P3 of about 0.4 mm, which is equal to the width Z of each of the ball pads 101-105 plus the interval of two adjacent ball pads of about 0.18-0.2 mm ( $P3=Z+V$ , where  $V=0.18\text{-}0.2\text{ mm}$ ). According to an embodi-

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ment of the invention, for example, the width Z of each of the ball pads **101-105** is 0.2 mm. The first row R1 and the second row R2 are arranged with pitch P2.

FIG. 3 also illustrates a basic pad unit PU consisting of five pads. For example, the basic pad unit PU1 is composed of two ball pads **101** and **102** in the first row R1 and three ball pads **201-203** in the second row R2, and the next basic pad unit PU2 is composed of two ball pads **103** and **104** in the first row R1 and three ball pads **204-206** in the second row R2. The array of the ball pads for the multi-row QFN is constructed by repeating the basic pad units PU. It is advantageous to use the present invention because the number of the ball pads can be increased by 4.5% for a 12.7 mm×12.7 mm dual-row QFN package (from 176 pins to 186 pins). Alternatively, the size of the dual-row QFN package can be reduced by 7.7% for a 176-pin dual-row QFN package (from 12.7 mm×12.7 mm to 12.2 mm×12.2 mm). Further, it is easier to assign the differential pair when routing the layout for the multi-row QFN packages.

Those skilled in the art will readily observe that numerous modifications and alterations of the device and method may be made while retaining the teachings of the invention. Accordingly, the above disclosure should be construed as limited only by the metes and bounds of the appended claims.

What is claimed is:

1. A board-level pad pattern, comprising:
  - a plurality of staggered ball pads disposed within a surface mount region for mounting a multi-row Quad Flat No Lead (QFN) package, wherein the plurality of staggered ball pads comprises first ball pads arranged in a first row and second ball pads arranged in a second row, wherein the first ball pads in the first row are arranged at two different pitches, and the second ball pads in the second row are arranged at a constant pitch; and
  - a basic pad unit consisting of five pads composed of two of the first ball pads in the first row and three of the second ball pads in the second row, wherein an array of the plurality of staggered ball pads within the surface mount region for the multi-row QFN is constructed by repeating the basic pad units.
2. The board-level pad pattern according to claim 1, wherein the plurality of staggered ball pads is copper pads.
3. The board-level pad pattern according to claim 1, wherein the plurality of staggered ball pads is non-solder mask defined (NSMD) pads.
4. The board-level pad pattern according to claim 1, wherein the first ball pads in the first row are aligned to one another along a first direction.
5. The board-level pad pattern according to claim 4, wherein the second ball pads in the second row are aligned to one another along the first direction.

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6. The board-level pad pattern according to claim 5, wherein the first and second ball pads are arranged in a staggered manner in a second direction that is orthogonal to the first direction.

7. The board-level pad pattern according to claim 1, wherein the first ball pads in the first row are arranged at a pitch of 0.5 mm and a pitch of 0.7 mm.

8. The board-level pad pattern according to claim 1, wherein the second ball pads in the second row are arranged at a pitch of 0.4 mm.

9. A board-level pad pattern, comprising:

- a plurality of staggered ball pads disposed within a surface mount region for mounting a multi-row Quad Flat No Lead (QFN) package, wherein the plurality of staggered ball pads comprises first ball pads arranged in a first row and second ball pads arranged in a second row, wherein the first ball pads in the first row are arranged at two different pitches P1 and P2, and the second ball pads in the second row are arranged at a constant pitch P3;

at least two differential pair traces disposed between adjacent two of the first ball pads arranged in a first row; and

- a basic pad unit consisting of five pads composed of two of the first ball pads in the first row and three of the second ball pads in the second row, wherein an array of the plurality of staggered ball pads within the surface mount region for the multi-row QFN is constructed by repeating the basic pad units.

10. The board-level pad pattern according to claim 9, wherein the plurality of staggered ball pads is copper pads.

11. The board-level pad pattern according to claim 9, wherein the plurality of staggered ball pads is non-solder mask defined (NSMD) pads.

12. The board-level pad pattern according to claim 9, wherein the first ball pads in the first row are aligned to one another along a first direction.

13. The board-level pad pattern according to claim 12, wherein the second ball pads in the second row are aligned to one another along the first direction.

14. The board-level pad pattern according to claim 13, wherein the first and second ball pads are arranged in a staggered manner in a second direction that is orthogonal to the first direction.

15. The board-level pad pattern according to claim 9, wherein  $P1=X+2Y+Z$  and  $P2=2X+3Y+Z$ , wherein X is a width of each of the differential pair traces, Y is the space between the differential pair traces and the adjacent two of the first ball pads arranged in a first row, and Z is a width of each of the first ball pads.

16. The board-level pad pattern according to claim 9, wherein  $P1=0.5$  mm,  $P2=0.7$  mm, and  $P3=0.4$  mm.

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